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(54) **MARINA SLIP OCCUPANCY DETECTION**

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(57)

ABSTRACT

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Related U.S. Application Data

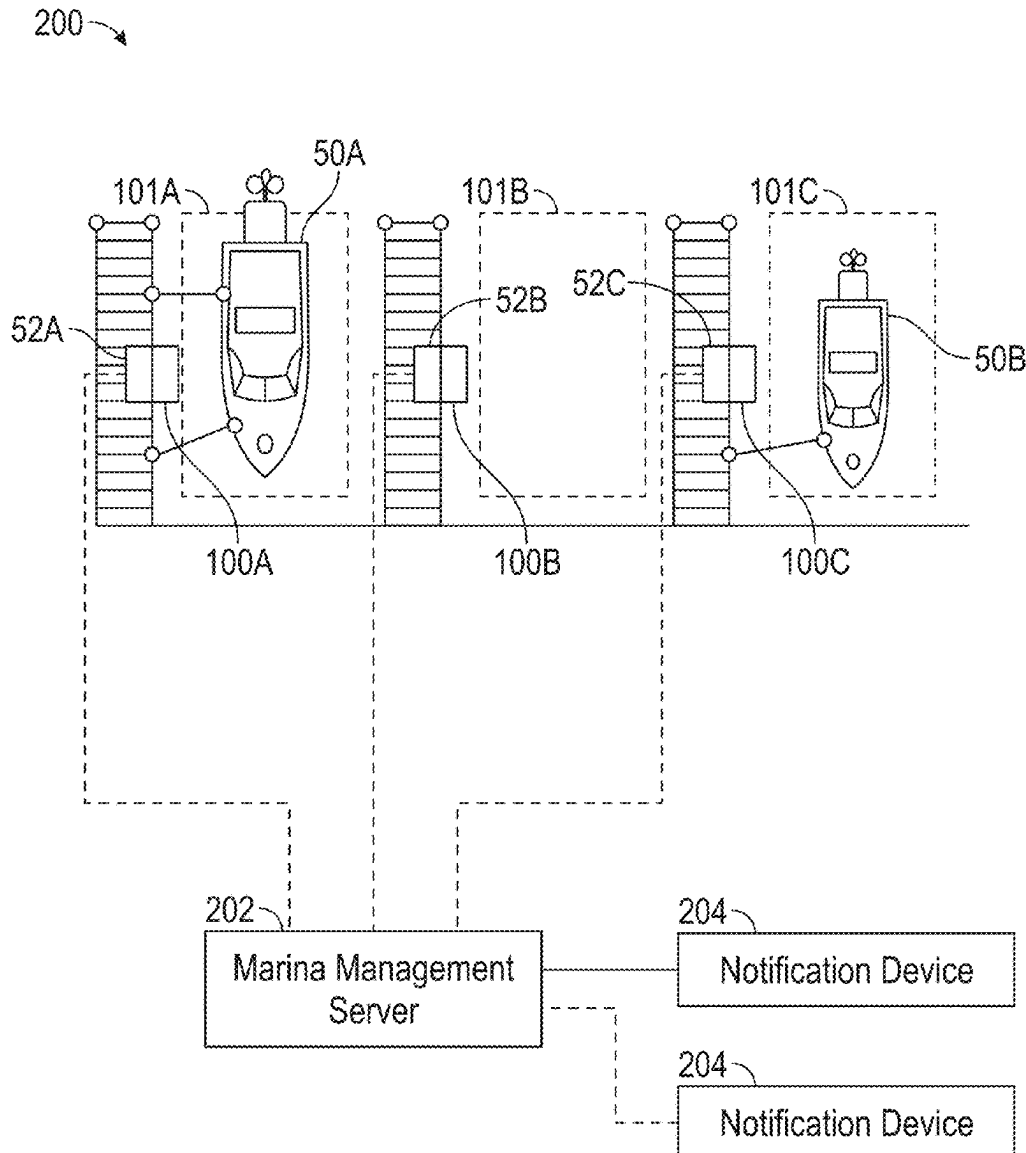
(63) Continuation of application No. 18/594,421, filed on Mar. 4, 2024, now Pat. No. 12,272,248.

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A marina slip occupancy detection system includes at least one watercraft detector configured to detect the presence and/or absence of a watercraft in a target area. A process of the watercraft detector is configured to communicate with a remote server with indications of occupancy or occupancy change. The remote server is configured to notify a dock-hand, marina manager, and/or the watercraft owner/operator depending on the occupancy change and predetermined configurations of the remote server.



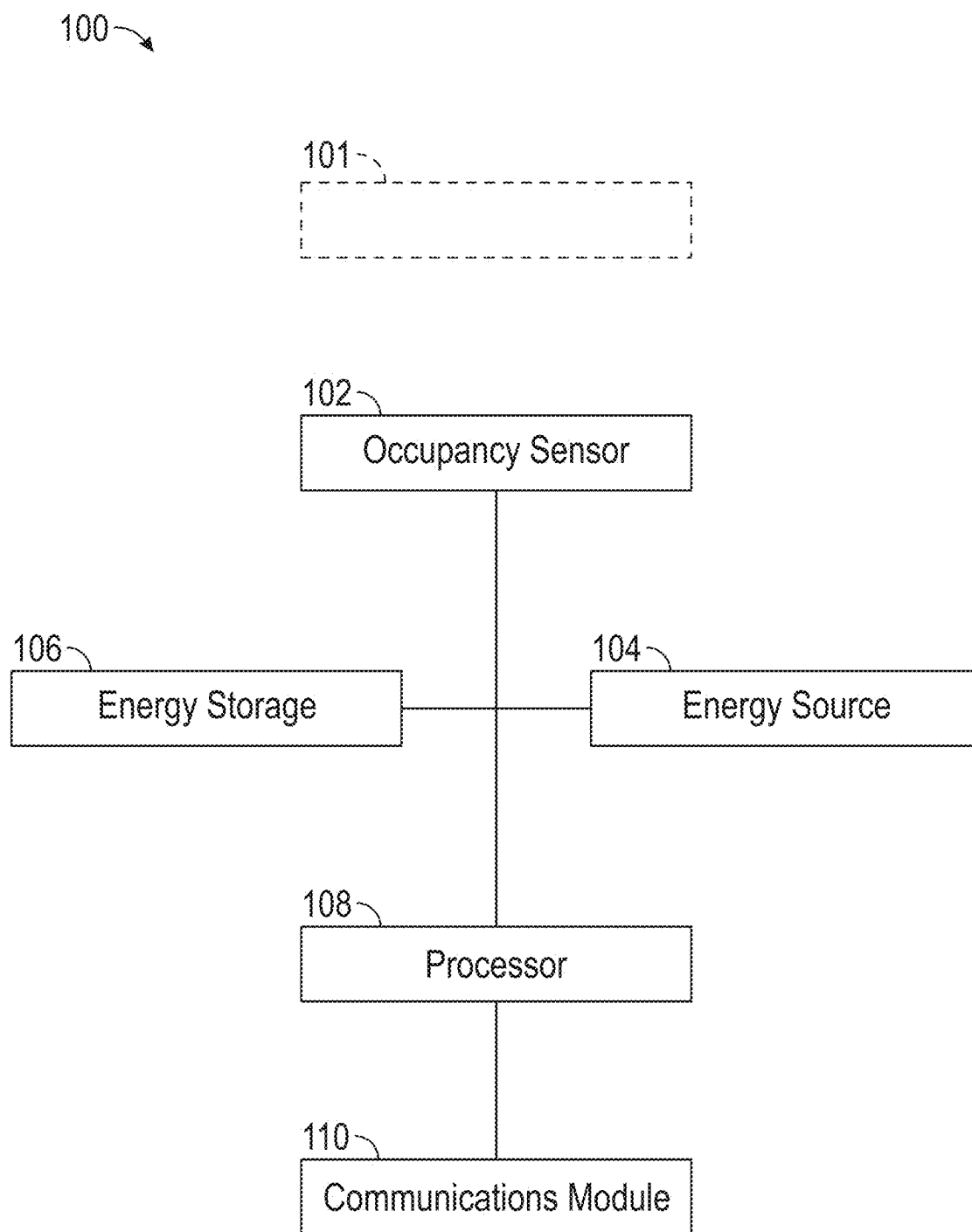


FIG. 1

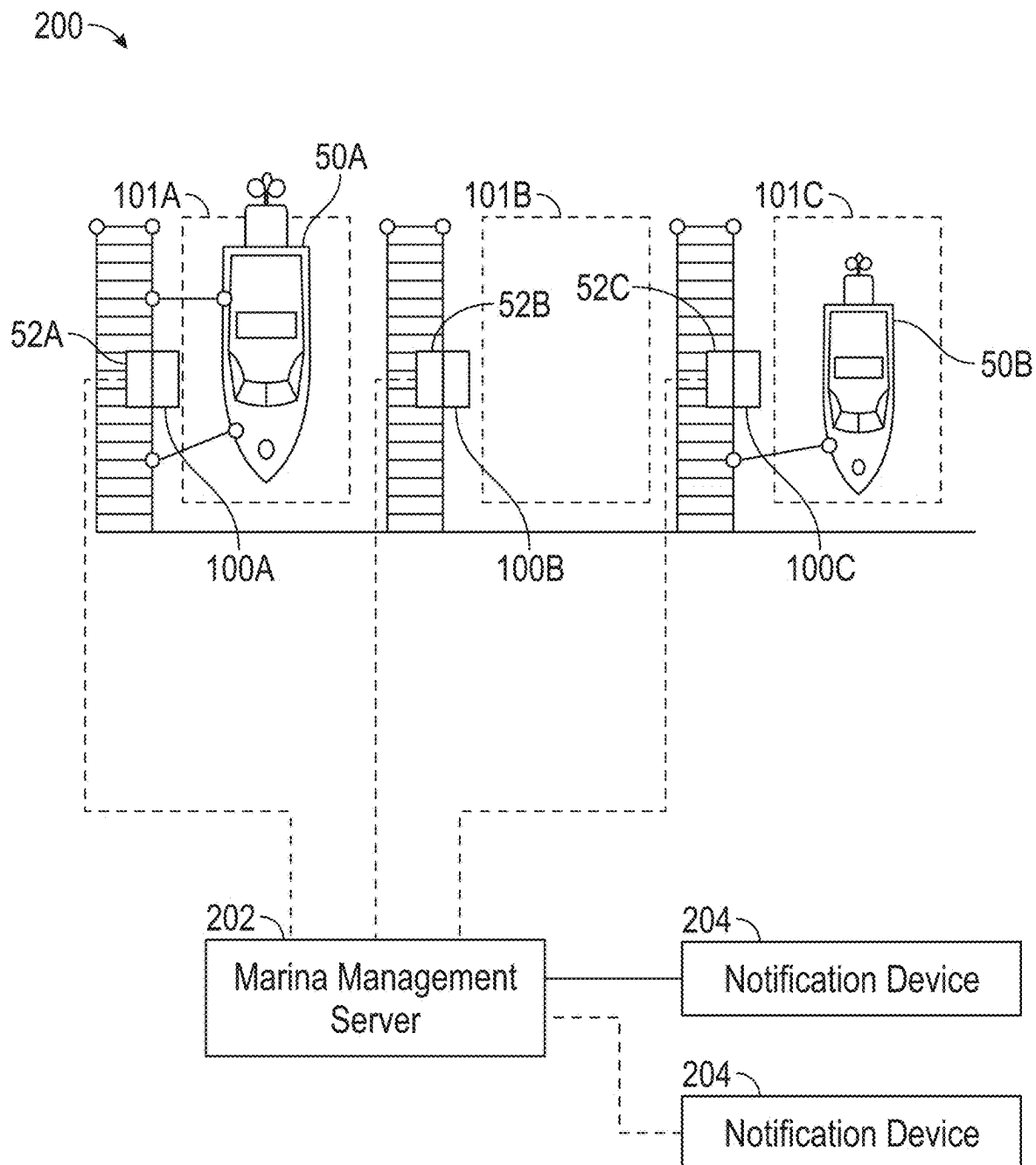


FIG. 2

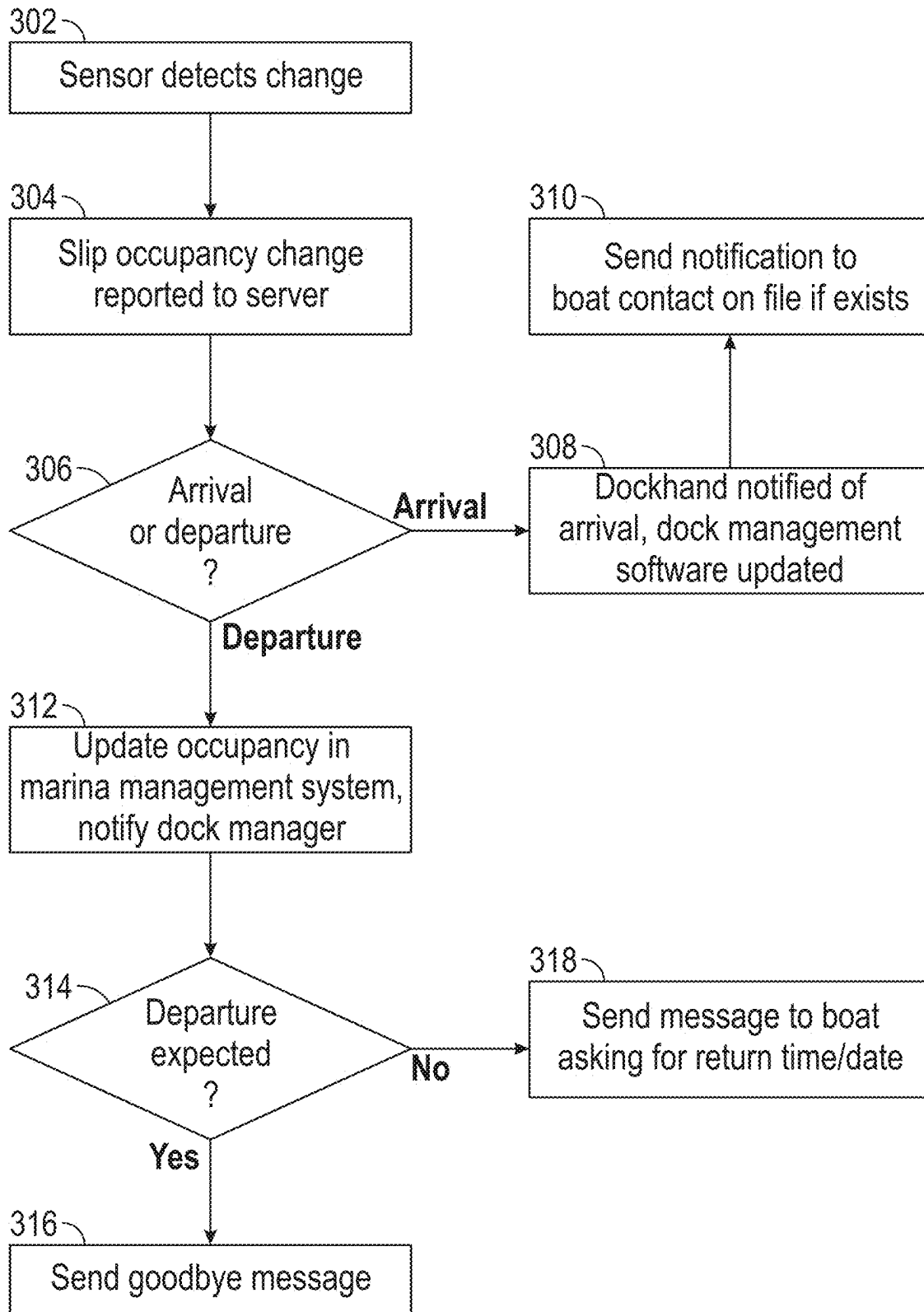


FIG. 3

MARINA SLIP OCCUPANCY DETECTION

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application is a continuation of U.S. patent application Ser. No. 18/594,421, filed Mar. 4, 2024, which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure generally relates to detecting slip occupancy and, in particular, to devices and systems for detecting and notifying marina management of occupancy of a marina slip for watercraft.

BACKGROUND

[0003] In the marina industry a significant revenue source are transient boat slips rental where the marina operation staff may re-rent a slip for one or more shorter duration stays when the longer duration renter is not currently using the slip. One problem marina operation staff often face is that they are not aware in a timely manner when these slips are no longer occupied and could be rented again. The long-term holders (e.g. seasonal or annual renters) do not always notify the marina office management when they are leaving and returning, thereby making the marina operation staff unaware of potential revenue opportunities. There are also cases where expected watercraft arrivals actually arrive at a much earlier or later time than expected and prompt notification to a dockhand to assist with docking is helpful to the watercraft operator. There is also a problem of slip theft where a watercraft may dock at an unoccupied slip and leave before it is noticed by the marina operation staff. The marina operation staff received no compensation for this dockage and may be unaware that the watercraft ever occupied the slip.

SUMMARY

[0004] The present disclosure advantageously provides devices, systems and methods for performing marina slip occupancy detection. The devices and systems disclosed herein can be arranged to detect when a watercraft (e.g. boat, jet-ski, etc.) arrives or leaves a marina slip. Timely notification of the arrival or departure of a watercraft from monitored slips at the marina allows for marina operation staff to more effectively operate the marina. For example, timely notification of the arrival or departure of watercraft allows for marina operation staff to advertise to non-docked watercraft that space is available for docking, be more promptly aware when slip theft is occurring, and more promptly alert dockhands (part of marina operation staff) to attend to arriving watercraft belonging to long-term renters of the marina.

[0005] In some embodiments, a marina slip occupancy detection system includes a remote server and a watercraft detector. The watercraft detector includes an occupancy sensor configured to detect a presence and/or an absence of a watercraft in a target area and to generate an occupancy signal based on the detected presence and/or the detected absence. The watercraft detector further includes a processor configured to receive the occupancy signal from the occupancy sensor. The processor is configured to communicate with the remote server to provide an occupancy indication of a slip associated with watercraft detector.

[0006] In some embodiments, the marina slip occupancy detection system includes a plurality of watercraft detectors configured to detect the presence and/or absence of a watercraft in a respective target area or slip. The processor of each of the plurality of watercraft detectors is configured to communicate with the remote server to provide an occupancy indication of each respective target area or slip.

[0007] In some embodiments, a method of monitoring marina slip occupancy includes detecting, with at least one watercraft detector, an occupancy of a plurality of slips for watercraft; communicating, by the at least one watercraft detector, with a remote server to provide an occupancy indication of each slip of the plurality of slips; and communicating, by the remote server, with a notification device to indicate a change in occupancy status of a slip of the plurality of slips.

[0008] Objects, features and advantages of the present invention will become apparent in light of the description of embodiments and features thereof, as enhanced by the accompanying figures.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is a schematic diagram of a watercraft occupancy detector in accordance with the present disclosure;

[0010] FIG. 2 is a schematic diagram of a marina slip occupancy detection system in accordance with the present disclosure; and

[0011] FIG. 3 is a flow diagram of a marina slip occupancy detection method in accordance with the present disclosure.

DETAILED DESCRIPTION

[0012] Referring to FIG. 1, a schematic diagram of an exemplary watercraft detector **100** is shown according to the present disclosure. The detector **100** includes an occupancy sensor **102**, an energy source **104** and/or an energy storage **106**, a processor **108** and a communications module **110**.

[0013] The occupancy sensor **102** can be configured in the form of any known occupancy detector. For example, the occupancy sensor **102** may be a passive infrared (PIR) sensor, ultrasonic sensor, break beam sensor or camera. In some embodiments, the occupancy sensor **102** can be a power pedestal electricity usage/connection sensor that detects when a watercraft hooks up to and/or disconnects from a power pedestal connection associated with the slip. Whether integrated with a power pedestal or a standalone occupancy sensor unit, the occupancy sensor **102** is preferably weather-resistant or weatherproof such that the sensor **102** can be installed in outdoor marina environments, where harsh wind, rain and sunlight elements are often encountered.

[0014] The occupancy sensor **102** is configured to detect the presence and/or the absence of a watercraft in a target area **101**. The occupancy sensor **102** is further configured to generate an occupancy signal based on the detected presence and/or absence of a watercraft in the target area **101**.

[0015] The energy source **104** can be any source providing electric power, such as an electric grid or solar cell(s) or a hardwired data line. The energy storage **106** can be a battery, such as a lithium ion rechargeable battery. In some embodiments, the watercraft detector **100** operates solely off of energy source **104** power without an energy storage **106**. In some embodiments, the energy storage **106** is provided to

power the watercraft detector **100** during periods of interruption of electric service of the energy source **104**.

[0016] The processor **108** is configured to communicate with an external device or system through the communications module **110**. The communications module **110** is configured to communicate with external devices and/or systems through any known means of wired and/or wireless communication protocols. For example, the communications module **110** may be configured to communicate with a remote server through wi-fi, cell towers, radio networks, etc. In some embodiments, the communications module **110** is configured to communicate over a power source wire, or through blinking visible or non-visible lights. In some embodiments, the communication may be through a marina's existing Wi-Fi amenity that is offered to guests.

[0017] Referring to FIG. 2, a schematic diagram of an exemplary marina slip occupancy detection system **200** is shown according to the present disclosure. The system **200** includes three watercraft detectors **100A**, **100B**, **100C** (collectively “**100**”) each optionally associated and/or integrated with a dedicated power pedestal **52A**, **52B**, **52C**, a marina management server **202** and at least one notification device **204** configured to communicate with the marina management server **202** (or “remote server”). Each watercraft detector **100** is configured to detect the presence and/or absence of watercraft at respective target areas **101A**, **101B**, **101C** (collectively “**101**”). Each target area **101** is associated with an area where a watercraft can dock at a marina. While the shown system **200** is shown with three watercraft detectors **100** for detecting the occupancy at three slips, it should be understood that a system **200** according to the present disclosure can adapted based on the size and needs of any marina. Thus, the system **200** can be employed by any marina having one or more slips and, thus, one or more target areas **101** for occupancy detection.

[0018] In FIG. 2, a watercraft **50A** is docked in a first slip and within the target area **101A** monitored by the watercraft detector **100A**. No watercraft is docked in a second slip or within the target area **101B** monitored by the watercraft detector **100B**. Another watercraft **50B** is docked in a third slip and within the target area **101C** monitored by the watercraft detector **100C**. The processor **108** for each watercraft detector **100** communicates with the remote server **202** to provide an occupancy indication of the target area **101** or slip the respective watercraft detector **100** is associated with. The marina management server **202** communicates with the notification device(s) **204** in real-time to provide alerts, statuses and/or status changes of each slip.

[0019] In some embodiments, the processor **108** is configured to continuously provide the occupancy status of each slip by communicating the occupancy indication to the marina management server **202**, or periodically by periodically providing the status of the slip to the server **202** at specified intervals (e.g. every 1 minute, 5 minutes, or 10 minutes, etc.). In some embodiments, the processor **108** is configured to only communicate with the marina management server **202** when there is a change in status of the slip, i.e. a detected occupancy of the target area **101** changing from watercraft presence detected to watercraft absence detected or a detected occupancy of the target area changing from watercraft absence detected to watercraft presence detected.

[0020] Referring to FIG. 3, a flow diagram of an exemplary method of marina slip occupancy detection is shown

according to the present disclosure. The method beings at block **302**, where a watercraft detector **100** detects a change in occupancy, i.e. from detected presence to detected absence of watercraft or from detected absence to detected presence of watercraft. The method proceeds to block **304**, where the processor **108** of the watercraft detector **100** communicates with a remote server **202** of the occupancy change. Then, at block **306**, the remote server **202** determines whether the slip occupancy change involves a watercraft arrival or departure. If arrival, the remote server **202** at block **308** a dockhand is notified through a notification device **204**. Optionally, the remote server **202** can proceed to block **310** and send a communication to a boat contact on file, if one exists. For example, if the occupancy change of a slip belonging to a long-term renter, the remote server **202** can send a communication to the phone number, email address, or other contact means, that a dockhand is on their way to attend to the watercraft. Depending on the interaction level desired, the message could trigger interactive messaging with the boat owner/operator to trigger a workflow with marina management server (hosted on the remote server **202** or other server) via API calls. For example, the remote server **202** may be configured to generate and communicate a link to an online booking platform if boat information is known and not under contract with the marina operating entity.

[0021] If the remote server **202** determines that the occupancy change is a watercraft departure, the method proceeds to block **312** where the marina management system is updated and notification is provided to a dock manager through a notification device **204**. Optionally, the remote server **202** can proceed to block **314** and determine whether the watercraft departure was expected. If expected, the remote server **202** proceeds to send a goodbye message at block **316**. For example, the remote server **202** can send a communication to the contact information confirming departure of the watercraft. If not expected, the remote server **202** can at block **318** send a communication to the contact on file requesting a return date and/or time.

[0022] In some embodiments, a single watercraft detector **100** and/or occupancy sensor **102** is configured to monitor and detect occupancy of more than one target area **101**. For example, in some embodiments, a single camera occupancy sensor **102** may be able to view and detect the occupancy of multiple target areas associated with multiple slips. The processor associated with such a sensor monitoring more than one target area or slip is configured to include in the indication provided to the remote server **202** which target area or slip the occupancy indication is provided for. As such, the remote server **202** in such embodiments will not be configured to only interpret communications from one processor **108** as being only associated with one target area or slip of the marina.

[0023] In some embodiments, more than one watercraft detector **100** and/or occupancy sensor **102** may be arranged for monitoring or detecting the presence and/or absence of a watercraft in a single target area **101**. Additional detectors **100** and/or sensors **102** may be advantageous for providing redundancy or increasing confidence of occupancy detection. For example, a slip having more than one detector **100** or sensor **102** may advantageously be able to disregard false alarms or notifications that would be generated by a single detector **100** or sensor **102**. For example, the processor **108** or remote server **202** may be configured to not indicate a change in occupancy unless two or more detectors **100** or

sensors **102** were in agreement with their detections. The two or more detectors **100** or sensors **102** could be arranged spatially close to each other or instead spaced apart so that one detector **100** or sensor **102** is arranged substantially near where a front portion of a watercraft would be arranged when occupying the target area **101** or slip and the other detector **100** or sensor **102** is arranged substantially near where a rear portion of a watercraft would be arranged when occupying the target area or slip.

[0024] Advantageously, the devices, systems and methods of the present disclosure allow for optimized marina management by marina operation staff. The improved marina slip occupancy provided by the present disclosure provides for optimized outcomes that can lead to more effective service for long-term slip renters as well as increase revenue opportunities by making available slip rentals more timely notified to staff as well as more timely notification of individuals committing slip theft.

[0025] The system, computers, servers, devices and the like described herein may be any computer-based device having the necessary electronics, computer processing power, interfaces, memory, hardware, software, firmware, logic/state machines, databases, microprocessors, communication links, displays or other visual or audio interfaces, printing devices, and any other input/output interfaces, to provide the functions or achieve the results described herein. Computer or computer-based devices described herein may include any number of computing devices capable of performing the functions described herein, including but not limited to embedded computers, discrete processors, desktop computers, tablets, laptop computers, smartphones, smart televisions, and the like.

[0026] Although various features have been shown in different figures for simplicity, it should be readily apparent to one of skill in the art that various features may be combined without departing from the scope of the present disclosure.

[0027] The foregoing description of embodiments of the present invention has been presented for the purpose of illustration and description. It is not intended to be exhaustive or to limit the invention to the form disclosed. Obvious modifications and variations are possible in light of the above disclosure. The embodiments described were chosen to best illustrate the principles of the invention and practical applications thereof to enable one of ordinary skill in the art to utilize the invention in various embodiments and with various modifications as suited to the particular use contemplated.

1. A marina slip occupancy detection system comprising:
 - a remote server; and
 - a watercraft detector, the watercraft detector comprising:
 - an occupancy sensor configured to detect a presence and/or an absence of a watercraft in a target area and to generate an occupancy signal based on the detected presence and/or the detected absence; and
 - a processor configured to receive the occupancy signal from the occupancy sensor; and
 wherein the processor is configured to communicate with the remote server to provide an occupancy indication of a slip associated with watercraft detector;

wherein the remote server is configured to communicate with a notification device of marina operation staff to provide a notification to a user of a change in occupancy status of the slip.

2. The marina slip occupancy detection system according to claim 1, wherein the occupancy sensor is a passive infrared sensor, ultrasonic sensor or a break beam sensor.

3. The marina slip occupancy detection system according to claim 1, wherein the occupancy sensor is a camera.

4. The marina slip occupancy detection system according to claim 1, wherein the occupancy sensor is a power pedestal electricity usage/connection sensor.

5. The marina slip occupancy detection system according to claim 4, wherein the processor is configured to provide the occupancy indication as indicating occupied when the processor determines an electrical connection has been established with a power pedestal, and wherein the processor is configured to provide the occupancy indication as indicating unoccupied when the processor determines an electrical connection with the power pedestal has been disconnected.

6. The marina slip occupancy detection system according to claim 1, wherein the remote server is configured to transmit a message to a phone number or an email address of an owner or operator of a slip associated with the target area.

7. The marina slip occupancy detection system according to claim 1, wherein the remote server is configured to indicate to the notification device a slip status of each of the slips based on the occupancy indications provided by the processors of the plurality of watercraft detectors.

8. The marina slip occupancy detection system according to claim 1, wherein the processor is configured to continuously communicate with the remote server to provide the occupancy indication of the slip.

9. The marina slip occupancy detection system according to claim 1, wherein the processor is configured to periodically communicate with the remote server to provide the occupancy indication of the slip.

10. The marina slip occupancy detection system according to claim 1, wherein the processor is configured to communicate with the remote server to provide the occupancy indication of the slip only when the occupancy sensor detects a change in occupancy in the target area, the change in occupancy being:

a detected presence of watercraft in the target area changing to a detected absence of watercraft in the target area; or

a detected absence of watercraft in the target area changing to a detected presence of watercraft in the target area.

11. A marina slip occupancy detection system comprising:

- a remote server; and
- a plurality of watercraft detectors, each watercraft detector comprising:
 - an occupancy sensor configured to detect a presence and/or an absence of a watercraft in a target area and to generate an occupancy signal based on the detected presence and/or the detected absence; and
 - a processor configured to receive the occupancy signal from the occupancy sensor;

wherein the processor of each of the plurality of watercraft detectors is configured to communicate with the remote server to provide an occupancy

indication of a slip associated with the respective watercraft detector of the processor;
wherein the remote server is configured to communicate with a notification device of marina operation staff to provide a notification to a user of a change in occupancy status of the slip.

12. The marina slip occupancy detection system according to claim **11**, wherein the remote server is configured to indicate to the notification device a slip status of each of the slips based on the occupancy indications provided by the processors of the plurality of watercraft detectors.

13. The marina slip occupancy detection system according to claim **11**, wherein each of the plurality of watercraft detectors is integrated into a dedicated power pedestal.

14. A method of monitoring marina slip occupancy comprising:

detecting, with at least one watercraft detector, an occupancy of a plurality of slips for watercraft;
communicating, by the at least one watercraft detector, with a remote server to provide an occupancy indication of each slip of the plurality of slips;
communicating, by the remote server, with a notification device to indicate a change in occupancy status of a slip of the plurality of slips; and

notifying, with the notification device, a member of marina operation staff of the change in occupancy status of the slip.

15. The method according to claim **14**, further comprising generating, by the notification device, a visual and/or audible alert indicating the change in occupancy status of the slip.

16. The method according to claim **15**, wherein the notification device is desktop computer, a laptop computer, a smartphone, or a tablet.

17. The method according to claim **14**, further comprising transmitting, by the remote server, a message to a phone number or email address associated with the slip after the change in occupancy status.

18. The method according to claim **14**, wherein the change in occupancy of the status of the slip comprises a detected occupancy of a target area of the slip changing from watercraft absence detected to watercraft presence detected, and wherein no compensation has been provided to the marina operation staff for a watercraft occupying the slip.

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